

# Laboratório 7

Julia Maria Pereira Pinheiro

```
library(DBI)
library(RSQLite)
conn <- dbConnect(SQLite(), "database.sqlite3")
```

```
dbListTables(conn)
```

```
[1] "course_offerings"      "courses"              "grade_distributions"
[4] "instructors"          "rooms"                 "schedules"
[7] "sections"             "subject_memberships"  "subjects"
[10] "teachings"
```

2. Quem são e quantos são os professores que lecionaram disciplinas cujo tópico era estatística (subjects.abbreviation='STAT');

```
professores_stat <- dbGetQuery(conn, "
  SELECT DISTINCT
  i.name as nome_professor,
  i.id as id_professor
  FROM instructors as i
  INNER JOIN teachings as t ON i.id = t.instructor_id
  INNER JOIN sections as s ON t.section_uuid = s.uuid
  INNER JOIN subject_memberships as sm ON s.course_offering_uuid = sm.course_offering_uuid
  INNER JOIN (select * from subjects where subjects.abbreviation='STAT' ) as sub ON sm.subject_id = sub.id
  str(professores_stat)
```

'data.frame': 379 obs. of 2 variables:

```
$ nome_professor: chr "MINJING TAO" "DONALD PORTER" "SHENG WANG" "KUNLING HUANG" ...
$ id_professor : chr "4392906" "393380" "4146030" "5697552" ...
```

```
nrow(professores_stat)
```

```
[1] 379
```

3. O GPA americano é definido numa escala de 0 a 4, em que A = 4, AB = 3.5, B = 3, BC = 2.5, C = 2, D = 1 e F = 0. Determinando a nota média de cada oferecimento pela ponderação da quantidade de alunos em cada extrato com os valores numéricos de cada conceito, indique (no que se referente a disciplinas no assunto de estatística):

- Quem é o professor mais difícil?
- Quem é o professor mais fácil?

```
todos_professores <- dbGetQuery(conn, "SELECT
  i.id AS instructor_id,
  i.name AS instructor_name,
  SUM(gd.a_count + gd.ab_count + gd.b_count + gd.bc_count +
      gd.c_count + gd.d_count + gd.f_count) AS total_alunos,
  SUM(gd.a_count * 4.0 + gd.ab_count * 3.5 + gd.b_count * 3.0 +
      gd.bc_count * 2.5 + gd.c_count * 2.0 + gd.d_count * 1.0 + gd.f_count * 0.0) /
  NULLIF(SUM(gd.a_count + gd.ab_count + gd.b_count + gd.bc_count +
      gd.c_count + gd.d_count + gd.f_count), 0) AS gpa_medio
FROM instructors i
INNER JOIN teachings t ON i.id = t.instructor_id
INNER JOIN sections s ON t.section_uuid = s.uuid
INNER JOIN grade_distributions gd ON s.course_offering_uuid = gd.course_offering_uuid
      AND s.number = gd.section_number
GROUP BY i.id, i.name
HAVING SUM(gd.a_count + gd.ab_count + gd.b_count + gd.bc_count + gd.c_count + gd.d_count + gd.f_count) > 0
ORDER BY gpa_medio ASC")
str(todos_professores)
```

```
'data.frame':  12031 obs. of  4 variables:
 $ instructor_id  : chr  "3633324" "4286674" "3183375" "4284728" ...
 $ instructor_name: chr  "HOLLY CHEN" "FEI CHEN" "DAVID DYNERMAN" "JENNIFER LOSAW" ...
 $ total_alunos   : int   12  6  32  13  8  8  49  42  10  26 ...
 $ gpa_medio      : num   1.21  1.92  2.11  2.12  2.12 ...
```

```
head(todos_professores, 1)
```

```
  instructor_id instructor_name total_alunos gpa_medio
1      3633324      HOLLY CHEN           12  1.208333
```

```
tail(todos_professores, 1)
```

|       | instructor_id | instructor_name | total_alunos | gpa_medio |
|-------|---------------|-----------------|--------------|-----------|
| 12031 | 993578        | SABINE PELLETT  | 12           | 4         |

- Qual é a disciplina mais difícil?
- Qual é a disciplina mais fácil?

```
todos_cursos <- dbGetQuery(conn, "SELECT
  c.number AS codigo_disciplina,
  c.name AS nome_disciplina,
  SUM(gd.a_count + gd.ab_count + gd.b_count + gd.bc_count +
      gd.c_count + gd.d_count + gd.f_count) AS total_alunos,
  SUM(gd.a_count * 4.0 + gd.ab_count * 3.5 + gd.b_count * 3.0 +
      gd.bc_count * 2.5 + gd.c_count * 2.0 + gd.d_count * 1.0 + gd.f_count * 0.0) /
  NULLIF(SUM(gd.a_count + gd.ab_count + gd.b_count + gd.bc_count +
      gd.c_count + gd.d_count + gd.f_count), 0) AS gpa_medio
FROM courses c
INNER JOIN course_offerings co ON c.uuid = co.course_uuid
INNER JOIN grade_distributions gd ON co.uuid = gd.course_offering_uuid
GROUP BY c.number, c.name
HAVING SUM(gd.a_count + gd.ab_count + gd.b_count + gd.bc_count + gd.c_count + gd.d_count + gd.f_count) > 0
ORDER BY gpa_medio ASC")
str(todos_cursos)
```

```
'data.frame': 7365 obs. of 4 variables:
 $ codigo_disciplina: chr "636" "577" "321" "677" ...
 $ nome_disciplina : chr "Afro-American History Since 1900" "Blacks in Cities" "Applied Ma
 $ total_alunos : int 44 32 1096 24 16560 275 11 3450 1210 6 ...
 $ gpa_medio : num 2.4 2.41 2.44 2.46 2.51 ...
```

```
head(todos_cursos, 1)
```

|   | codigo_disciplina | nome_disciplina                  | total_alunos | gpa_medio |
|---|-------------------|----------------------------------|--------------|-----------|
| 1 | 636               | Afro-American History Since 1900 | 44           | 2.397727  |

```
tail(todos_cursos, 1)
```

|      | codigo_disciplina | nome_disciplina                     | total_alunos |
|------|-------------------|-------------------------------------|--------------|
| 7365 | 999               | Reading and Research-Accounting PhD | 6            |
|      | gpa_medio         |                                     |              |
| 7365 | 4                 |                                     |              |

```
dbDisconnect(conn)
```

```
Sys.setenv(TZ = "America/Sao_Paulo")  
Sys.time()
```

```
[1] "2026-09-23 19:25:42 -03"
```